

Project Design Phase Solution Architecture

Date	18 July 2026
Team ID	SL-TEAM-01
Project Name	Smart Lender – AI Loan Approval System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

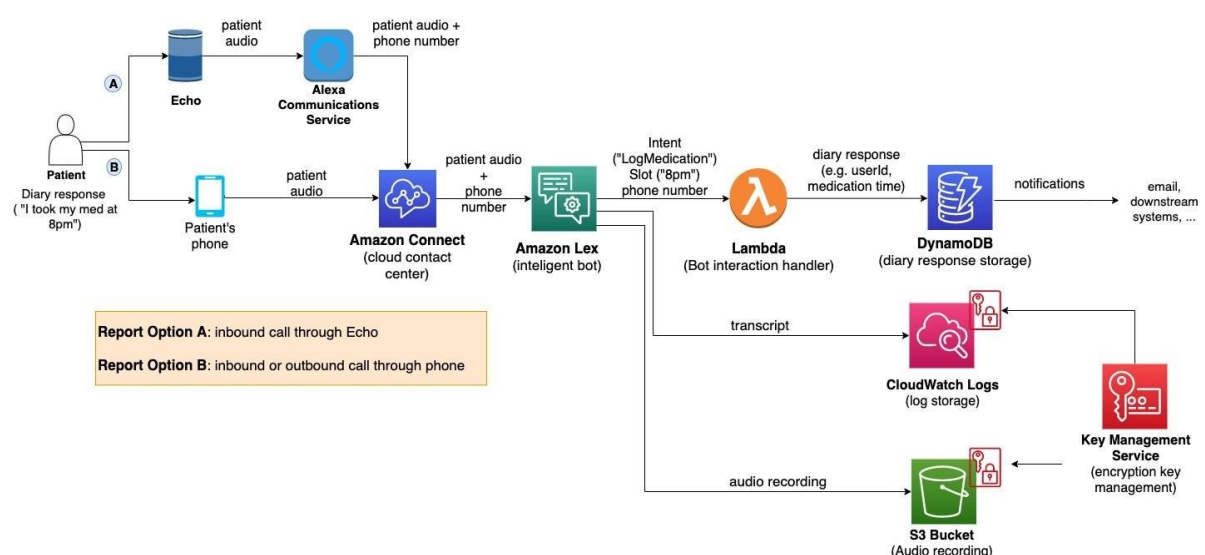


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

Smart Lender – Solution Architecture (Summary)

Architecture Flow:

- Presentation Layer: Single index.html page (HTML5 + CSS3) with sections for single application form, batch CSV upload, and results.
- Client Logic Layer: JavaScript functions handle validation (handlePrediction), scoring (simulatePrediction), result rendering (displayResult) and CSV batch handling (parseCSV, handleBatchPrediction, displayBatchResults).
- Rule-Based Scoring Engine: Computes a weighted score from credit history, income-to-loan ratio, education, employment type, property area and dependents, then normalizes it to an approval probability.
- Output: Approved/Rejected decision, approval probability, confidence level and an application summary rendered back into the page (or as a results table for batch mode).
- Future Layer (planned): Backend API + database + trained ML model to replace/augment the rule-based engine, enabling persistence, authentication and analytics.